VinBigData Chest X-ray Abnormalities Detection Dataset

Dataset Overview

The VinBigData Chest X-ray Abnormalities Detection competition dataset contains 18,000 chest X-ray images annotated for 14 thoracic abnormality types. Collected from two major Vietnamese hospitals (Hospital 108 and Hanoi Medical University Hospital), it addresses the lack of large-scale, high-quality annotated datasets for chest X-ray analysis in Southeast Asia. Each image has multi-label bounding box annotations from 17 board-certified radiologists, making it uniquely valuable for robust model training despite label noise.github+1

Data Files and Structure

The dataset includes DICOM images and CSV files:

- **train.dicom**: 15,000 training images.
- **test.dicom**: 3,000 test images.
- **train.csv**: Contains image_id, class_id, rad_id, x_min, y_min, x_max, y_max for training annotations.
- **sample_submission.csv**: Template for predictions.kaggle+1

Images are frontal/lateral PA/AP views in DICOM format (~1MB each), with pixel spacing around 0.168 x 0.168 mm. Related resized PNG datasets exist for faster processing (512px or 1024px).kaggle+2

Abnormality Classes

14 classes cover common pathologies:

Class ID	Name	Description Example
1	Aortic enlargement	Widened aortic knob
2	Atelectasis	Lung collapse
3	Calcification	Vascular deposits
4	Cardiomegaly	Enlarged heart
5	Consolidation	Pneumonia filling
6	ILD	Interstitial lung disease
7	Infiltration	Diffuse opacity
8	Lung Opacity	General haze
9	Nodule/Mass	Lesions >3cm
10	Other lesion	Unspecified
11	Pleural effusion	Fluid in pleura
12	Pleural thickening	Scar tissue

Class ID	Name	Description Example
13	Pneumothorax	Collapsed lung
14	No finding	Normal study

Annotation Process

Annotations were crowdsourced from 17 radiologists via VinLab, with each image labeled independently by 3-4 experts. This introduces label variability, beneficial for training robust detectors but requiring techniques like label smoothing or ensemble labeling. The dataset paper "VinDr-CXR" details inter-observer agreement metrics.github+2

Key Statistics

- Total images: 18,000 (83% train, 17% test).
- Abnormal images in train: ~29% (4,400+ with ≥1 bbox).
- Total bounding boxes: ~22,000 across classes.
- Most common: Aortic enlargement, cardiomegaly; rarest: Pneumothorax.github+1 Class imbalance is severe, with "No finding" dominant and pneumothorax under 1%.

Data Quality and Challenges

Challenges include:

- **Label noise**: Multi-rater disagreement (e.g., 20-30% for subtle findings).
- **Class imbalance**: Requires focal loss or oversampling.
- **Box quality**: Some imprecise due to radiologist speed.
- **DICOM handling**: Windowing needed for visualization (Lung: -500 to 1500 HU). EDA notebooks highlight duplicate rad_id patterns and image metadata variations.

Usage Guidelines

1. Convert DICOM to PNG using pydicom/pillow.
2. Apply CLAHE for contrast enhancement.
3. Use EfficientDet/YOLOv5 for detection; handle "14" class specially.
4. Post-process: NMS at IoU 0.5, confidence >0.3.github+1 Pseudocode for loading: `df = pd.read_csv('train.csv'); groupby image_id for bboxes`.

Model Evaluation

mAP@IoU=0.4; aggregate per-class AP then mean. Top competition scores ~0.25 public LB; handle class 14 carefully (dummy box if max conf <1).vinbigdata+1

Research Applications

Ideal for transfer learning on NIH ChestX-ray14 or MIMIC-CXR; fine-tune for Vietnamese demographics. Extensions include segmentation via Mask R-CNN.kaggle+1

Competition Insights

1,277 teams; winner used TTA, SNAP ensemble, custom loss for noise. Post-competition, dataset used in papers on noisy label learning.institute.vinbigdata+1